CAIRN: Ownership Obstruction and Evaluation Design in Higher-Order Models of Oral History

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***Abstract*—A rank ladder is an information granulation, and combinatorial complexes make one computable, with a growing family of topological networks passing messages along it [1, 2]. Networks in that family act on cells, each a subset of a ground set, so where that set holds entities and the items above them are singly owned, two items with one owner are one cell. Singly owned items are what an oral-history archive holds: across 526 interviews, 13,315 described moments realise 455 supports over 470 narrators, so no combinatorial complex over those narrators holds them apart, a collapse we call the *ownership obstruction*. Lifting from rank 0 carries it into the features: at most 455 distinct rank-1 inputs, a 29.3-fold collapse. The split decides whether the collapse is visible. 64.8% of narrators straddle a random split, a rate the entity-degree distribution predicts at 64.1% before any model exists. Given that exposure, a complex network appears to beat a lossless typed-star expansion 0.388 to 0.106 in T1 MAP, and trails it 0.098 to 0.139 once narrators are held out, while destroying entity identity removes the complex's gain. Gains of that kind track the inputs rather than the operator: the star leads as registered, carrying spectral encodings the complex never receives, the ordering reverses once both read the same features, and on those matched inputs both architectures leak, 0.297 against 0.177. Assigning the middle rank is itself underdetermined, three programmatic operationalisations of one written manual agreeing at** α = 0.039**. What is new is the combination: an injectivity test that runs before training, an exposure rate predicted in closed form, and a controlled ranking reversal inside an owner-grounded hierarchy. The magnitude of the reversal is a finding about this archive; the obstruction itself is structural, and reaches any collection whose items are singly owned.**[1]

***Index Terms*—information granulation, granular computing, topological deep learning, higher-order networks, evaluation design, data leakage, oral history archives**

I. Introduction

A combinatorial complex assigns each cell a rank, so that each rank holds a coarser information granule than the one below, and lets messages travel between ranks as well as within them. Across those ranks, the operator family is rank-aware in a way graph and hypergraph message passing is not, and the accompanying networks are designed to exploit that [1, 2]. For an archive, the appeal of such networks is immediate, because testimony has obvious levels: a narrator, a moment inside an interview, an event that several narrators describe, a longer episode that contains those events. Those levels are tempting to read as a rank ladder, and reading them that way meets a counting problem (Figure 1): a combinatorial complex identifies a cell with the ground elements it contains, so two moments told by one narrator are one cell, and this archive's 13,315 moments carry 455 distinct supports over 470 narrators. Those collapsed supports are what we call the *ownership obstruction*.

Before a rank-structured model can be said to help, three things have to be true.

P1. Rank carries information beyond size. Coarsening a partition ordinarily produces larger blocks, so rank and size are expected to move together in any granulation. What a curated ladder has to add to any granulation is the residual: if rank is recoverable from cell size alone, the ladder is a sorting of set sizes and nothing is gained by calling it structure.

P2. The specification is stable. If faithful independent implementations of the same written manual assign a term to different ranks, the ladder cannot be operationalised from the manual alone, and what fills the gap is the judgement of whoever built it.

P3. The structure pays. If a typed star expansion matches a full complex at equal parameter count, the complex has not earned its additional structure.

We fixed thresholds and failure outcomes for all three before looking at any model result. Against them, on an archive of 526 interviews and 13,315 described segments, **rank carries information beyond size, the specification moves when it is applied twice, and the typed star beats the complex at equal parameter count**. The induced partition moves under a defensible change of granularity as well, and the star's margin comes from its inputs: on matched features the ordering reverses. That ladder is real; the machinery built on it has still to earn its place.

A fourth result, undeclared in advance, matters most and concerns the train/test split. Under a random split the ordering of architectures reverses and the complex's advantage is large, while it leads under no other design axis we varied, which places that advantage in the split rather than in the operator.

The reason lies in the feature construction these models are normally given. A cell's features are aggregated from its constituent rank-0 narrators, and nearly every moment has a single narrator, so such a moment's input vector *is* its narrator's vector, and two such moments land on both sides of a random split as the same vector. Shuffling narrator identity removes that vector's privilege and with it the complex's entire random-split advantage, while replacing the moment features with passage-specific text does not, which locates the leak in the narrator layer. Those extra ranks buy a shorter path to who is speaking.

[1] Data, code and audit record: https://github.com/AlexDongzeyu/CARIN.

Three things are new here. First, an exact support-injectivity test decidable on the encoding before a model exists, where the surrounding literature offers diagnostics that correlate with downstream behaviour. Second, an entity-exposure rate the degree distribution predicts in closed form, so the leak is quantifiable from the data alone. Third, a controlled reversal of the architecture ranking inside an owner-grounded hierarchy, with the mechanism named and intervened on. All three describe an encoding and a precondition on it, one level below the architecture comparisons they constrain.

II. Archive and rank ladder

Our corpus is the Densho Digital Repository's Japanese American oral-history collection: 526 interviews covering 470 narrators, 13,315 segments carrying archival description, and 519 transcripts, at a median of 23 segments per interview. A segment enters the event ranks only if it carries a topic term, which 4,600 of them do. Those ranks are taken from the archive's own descriptive practice: rank 0 a narrator, rank 1 a described moment within an interview, rank 2 an event or site, rank 3 the episode containing it.

That ladder cannot be a combinatorial complex over the narrators, and the reason is counting.

Proposition 1 (Ownership obstruction): Let S be a ground set and let a collection C_1 carry a support map $\mathrm{supp} : C_1 \to \mathcal{P}(S) \setminus \{\emptyset\}$. The cells of a combinatorial complex over S are subsets of S, so any two cells with the same support are the same cell. If C_1 realises fewer than $|C_1|$ distinct supports then supp is not injective, and no combinatorial complex over S carries C_1 as distinct cells. Where every cell has a single owner in S, that happens as soon as $|C_1| > |S|$.

The archive meets the condition at rank 1 (Figure 1b). Over 470 narrators at that rank, 13,315 described moments realise 455 distinct supports, one of them shared by 142 moments, and the counts repeat in all 9 design cells of the sweep because rank-1 cells are the archive's segments. Among those supports, 441 name single narrators, which is what holds the cell count below 470, and the remaining 14 are the sets of two to four speakers who share a segment. Ranks 2 and 3 are injective, so the obstruction sits at the rank where moments live. What we build at that rank is a *named complex*: cells carry identifiers, the incidence matrices are indexed by those identifiers, and the operators are well defined on them.

Three encodings escape the obstruction. Putting moment identifiers into the ground set restores injectivity, at the price that rank 0 stops meaning a narrator and the archive's narrator-to-moment-to-event-to-episode reading goes with it. Giving moments their own object type also works, but the ladder is then no longer a single complex. Treating it as a general ranked incidence structure works as well, and is the route taken here.

Only the rank-2 equivalence relation is varied. Three granularities of that relation are defined in advance: *coarse* merges a topic leaf into its parent path, *mid* uses the leaf as given, and *fine* splits a leaf by the geography attached to the segment. Mid, the leaf as given, is the registered primary. No archival principle makes mid primary, so a conclusion that depends on the choice is a conclusion about us.

The registration defined the two finer levels with a *chronology* decade bin and the implementation splits by geography instead: Densho segments carry no date field, so the decade bin needed an undeclared date-inference step with its own error rate. Auditing the registration against the code surfaced that after the results were in. Those results therefore describe sensitivity to a topical and geographic equivalence relation rather than to temporal binning, and no claim of temporal invariance appears anywhere below.

Transcript speaker labels are recovered before any of this, because attributing testimony to the wrong speaker would corrupt every rank-0 incidence. Archive labels name the speaker for 77.7% of 220,471 turns, and a rule-based classifier reproduces them with accuracy 0.806, and cleaning removes 13.2% of characters.

A. Tasks and models

Two tasks are scored, chosen because they ask opposite things of a representation.

T1, corroboration retrieval. Given a held-out passage by narrator n, rank all other passages in the archive so that passages by *different* narrators describing the same rank-2 event come first. This is an archive-motivated retrieval task: find me another account of this. Two design choices keep that retrieval non-trivial. Both concern the query: its own cell membership is masked, so the answer cannot be read off the structure the model is handed, and any candidate sharing a narrator with it is excluded from the positives, so a model cannot score by recognising the speaker instead of the event. We score MAP over 400 held-out queries, with nDCG@10, Recall@50 and MRR alongside.

T2, incidence prediction. Given a narrator and an event cell, predict whether that narrator is incident to that event, a link task over the bipartite incidence relation, scored by AUC at a declared ratio of ten sampled negatives per positive. In the primary regime, negatives come from cells at a short random-walk distance, so they are plausible.

By design, the two tasks are not aligned. T1 is scored over moment representations and T2 over narrator–event pairs, so a representation can be good at one and poor at the other, an asymmetry the ablations in Section VI-C turn into the paper's sharpest result.

Models. All learned models share the same frozen sentence encoder and a common budget of at most 132,219 parameters, with hidden width selected per model to meet it.

- **M0, MLP.** Per-cell feature transform with no message passing at all, the floor for "does any structure help".
- **M1, dense retrieval.** Cosine similarity over the frozen embeddings, with *zero* learned parameters, scored by identical metric code; if competitive, it bounds what any learned model can claim.
- **M2, untyped star.** Expansion of the complex to a bipartite graph, with a single weight matrix applied to every relation, discarding rank.

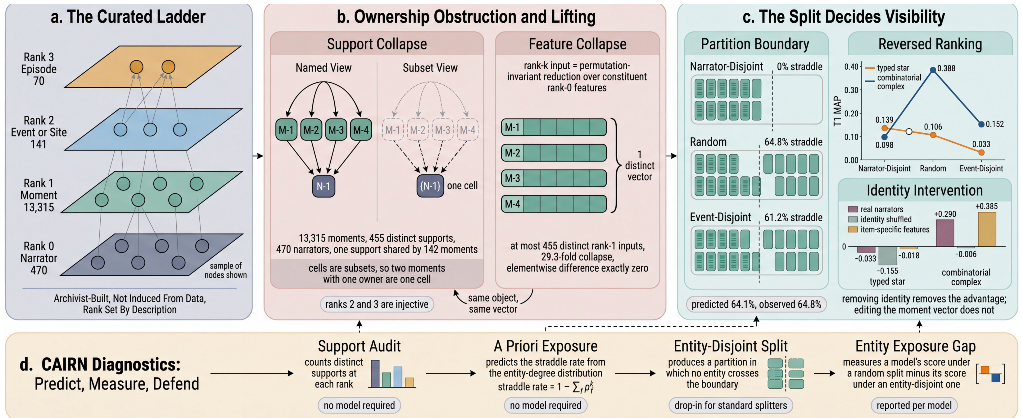

Fig. 1. The ownership obstruction and what it does to evaluation. **a**, The archive's four-level descriptive ladder; node counts are illustrative, tier counts exact. **b**, Rank-1 moments are almost all singly owned, so as subsets they are indistinguishable: 13,315 moments realise 455 supports over 470 narrators, leaving at most 455 distinct rank-1 inputs after lifting. **c**, Whether the collapse is visible depends on the partition. Straddle rate is the fraction of narrators on both sides; the predicted rate follows from the entity-degree distribution alone, with p_f the proportion of fold f and k the cells an entity owns. T1 MAP reverses under a random split, and shuffling narrator identity removes the complex's gain while the star's, never positive, falls further. **d**, The released diagnostics; the first two require no trained model.

- **M3, typed star.** Same expansion with relation-specific weights, so rank is available as a *type* but not as an operator. This model additionally receives spectral hypergraph encodings [3] that the complex does not.
- **M4, hypergraph baselines.** Rank-agnostic hypergraph models that treat events as hyperedges over narrators: AllSet-style attention [4], ED-HNN equivariant diffusion [5], and Hypergraph-MLP [6], which removes message passing entirely and is included as a skeptical control.
- **M5, combinatorial complex network.** Rank-aware message passing: per layer, up the ladder $r_0 \to r_1 \to r_2 \to r_3$, back down it, and within-rank exchange through shared cofaces, with rank-specific weights throughout. This is the model the study is about, a CCNN-style network over the named complex of Proposition 1 rather than over a standard combinatorial complex the archive does not admit; the operator family is that of Hajij et al. [1], indexed by cell identifiers.

M2 and M3 are the load-bearing comparisons. Both see the same incidence structure as M5, and M2 differs from it only in whether rank is an operator or a label, so it isolates that question; M3 additionally receives the spectral block, so a margin it wins answers whether a strong simpler baseline beats the complex. Both comparisons appear in Section VI-C, with an input-matched M3 that isolates the operator.

Implementation. Every learned model runs two message-passing layers with ReLU and dropout 0.1 over one objective, binary cross-entropy on incidences against ten sampled negatives per positive, optimised with AdamW at learning rate 5×10^{-3}, weight decay 10^{-4}, for at most 800 epochs, stopping once validation AUC has not improved for 120; the best validation checkpoint is restored before anything is scored. Neighbourhood operators are the incidence matrices between consecutive ranks together with their transposes, plus within-rank exchange through shared cofaces. With those exchange operators fixed in advance, no hyperparameter search enters the primary grid: depth, learning rate, dropout and the negative ratio are fixed at these values for every model, and Section VI-C reports a separate equal-budget sweep over the two models the argument rests on. Across those models only one quantity varies, hidden width, binary-searched over $[8, 256]$ so that each lands within 15% of the shared 132,219-parameter budget. Without that budget, relation-specific models carry roughly twice the parameters of shared-weight models at equal width, which would leave any difference between them unattributable.

III. Related work

Granular computing studies description at multiple levels of resolution [7, 8]. Its standard constructions form granules from the data: partition hierarchies induced by indiscernibility relations [9], and granule boundaries chosen to trade coverage against specificity [10]. Against those constructions, the ladder examined here is a multilevel granular structure an institution already maintains, with levels named by archivists rather than induced from data. Naming the levels settles the usual question of how to form a granule and raises a different one, which this paper answers: three operational preconditions for deciding whether a curated hierarchical granulation can be realised and evaluated as a higher-order learning structure at all. Topological and higher-order deep learning proposes that data with multi-way structure should be modelled on complexes rather than graphs, and combinatorial complex neural networks [1] generalise the graph, simplicial and hypergraph cases under one operator family. By contrast, the hypergraph line is older and better characterised: spectral clustering on hypergraphs [3], then neural formulations [4, 5] that treat a hyperedge as a multiset to be aggregated.

For this paper, the skeptical evidence matters more than the enthusiastic. Tang et al. [6] show that removing message passing from a hypergraph network entirely costs little on standard benchmarks. Papillon et al. [11] report that generalised complex networks match or outperform their predecessors once architectures are matched, building them on Hasse-graph expansions, which makes our typed star a stripped-down member of the family they found strong. On a manifold benchmark, Schmidt et al. [12] find outcomes turn on representation and feature assignment rather than the operator. Concurrently, Papillon et al. [13] argue that the lifting step deserves inspection before training, and supply diagnostics whose metrics correlate with downstream performance. The diagnostic offered here is of a different kind: support injectivity is an exact property of the encoding, decidable before a model is fitted, and Proposition 1 says what its failure costs. None of that work names a mechanism; we name one and show it reverses an architecture ranking through the *evaluation design* alone. Rieck [14] argues the field should source natively higher-order data rather than keep lifting graph datasets; this archive is such a source, its ladder recorded by archivists, so the obstruction is a fact about the data rather than a preprocessing choice.

A longer methodological literature stands behind that. Retrieval and recommendation have repeatedly found that weak baselines manufacture apparent progress [15], and that train/test leakage inflates results in ways architecture ablations cannot detect [16]. For graph networks the split already reorders architectures [17]; we add the structure that makes the reordering available and the mechanism that carries it, which Section VIII measures. The corroboration retrieval task of Section II-A is a computational form of a question historians already ask, and our labelling apparatus, chance-corrected agreement [18] and its known paradoxes [19], comes from content analysis.

IV. Event layer and its extractor

Our primary analysis takes rank-2 cells from the archive's controlled vocabulary. Separately we built an automatic extractor for the case of an archive without one, and measured it: at the selected threshold of 0.60 it reaches an LEA F_1 [20] of 0.196 over 199 clusters, wrongly unifies 84.1% of comparable gold pairs, splits 81.0% of gold clusters, and leaves a false singleton rate of 0.367, above the ceiling fixed in advance. This is a declared kill and we treat it as one, and its scope is narrow: these rates bound the pipeline an archive *without* a controlled vocabulary would have to build. The release carries the error decomposition in the style of Kummerfeld and Klein [21]. A separate flagger marks 21 terms as contested, and we re-run the model grid with those terms removed to separate a method that does not work from a labelling that is disputed.

V. The pre-registered protocol

In advance, the registration names the primary cell, the metrics, the seed count and a kill threshold for each precondition. Each carries a substitution record, six in all: most consequentially no second human annotator and no expert panel, so the agreement study runs three programmatic protocols implementing the written manual.

That substitution changes what P2 can claim. Treating the programmatic protocols as a lower bound on human agreement would assume misreading is the only source of human disagreement, when two people who both know Minidoka is a camp would agree on its rank whatever the manual said. What the manual said does not fix the direction of that bias, so we claim none. What we report is narrower: three faithful implementations of one written manual compute different functions, which is evidence about the specification rather than about people, and it is what the question needs—can this ladder be assigned without residual judgement.

All models are held to the same parameter budget. In the primary cell the trained models span 2.8% between largest and smallest, every result in that grid is averaged over 10 seeds, and the text encoder is frozen and identical [22]. The factorial and checkpoint sweeps of Section VI-C run 5 seeds and the tuning grid 3, stated where each appears. Encoder aside, inputs are deliberately *not* matched: the typed star also receives spectral hypergraph encodings [3] that the complex does not, favouring the baseline by a choice fixed in advance. Section VI-C reports the matched variant that separates the handicap from the typing. Cost is not the obstacle: building the complex takes 0.23 s, features 0.66 s and the star expansion 0.20 s, against 10.7 s and 251 MiB for one training seed.

VI. Empirical results

A. Rank against cardinality

Before any fitting, the criterion was that a Spearman correlation above 0.9 between a cell's rank and its size would mean the ladder is cardinality in disguise. Observed size correlation is $\rho = 0.471$ at the primary granularity, and between 0.34 and 0.48 across every granularity and rank map we ran, so the criterion passes everywhere and P1 holds.

That criterion is a proxy for the claim, and it leaves open whether some nonlinear function of size recovers rank anyway, so we also ran the direct test on the rank-2 and rank-3 boundary, the one place annotators actually disagree. Predicting rank from log cell size across those 211 boundary cells, with five-fold cross-validation, gives balanced accuracy 0.511 for logistic regression and 0.500 for a depth-three tree, against a prior-only baseline of 0.500. The tree's cross-entropy is worse than the prior's, and a k-nearest-neighbour estimate of the mutual information between size and that distinction returns 0.000 nats, the floor the estimator clips a non-positive value to. Over all nine granularity and rank-map cells the best any size-only predictor reaches is 0.611, so P1 no longer rests on a correlation threshold.

One scope note belongs here: coarser granules are usually larger, and nothing in granular computing forbids that. P1 is a precondition on *this* construction, where rank records descriptive abstraction rather than headcount, so a ladder recoverable from size would carry no information the cardinality did not already carry.

Two conditions bound it. Ranks 0 and 1 are size-constrained by construction, since a segment has one speaker, so pairs involving them re-measure that definition rather than the ladder; over all pairs at ranks 1–3 the ladder orders by size correctly 81.4% of the time and inverts 1.0% of the time. At the rank-2/rank-3 boundary, the one the annotators dispute, the picture depends on granularity: 37.1% of those 9,870 pairs invert at mid, rising above half at coarse and falling to about a tenth at fine. P1 passes, and that boundary is where it is closest, so how often the ranks invert is not a stable property of the archive.

A star expansion is also lossless, which is what licenses the comparison.

Proposition 2 (Star expansion is lossless): Let K be a named complex whose cells each carry a rank and a member set. Its typed star expansion places one node per cell and one typed edge per membership, with the edge type recording the rank. Rank and membership are recoverable from the expansion, so K is determined by its typed star.

Round-tripping all 13,996 cells recovers every rank and member set, with no cell missing, added or altered. The star also receives richer cell features, a handicap chosen in advance and described in Section II-A, which makes a negative result conservative. Being lossless, the star carries the same incidence information arranged more simply, so any advantage the complex shows has to come from the operator.

B. Latitude left to the implementer

Three protocols implementing the same written manual assign ranks to 211 terms. After a documented adjudication round, their rank assignments agree at $\alpha = 0.039$, against a floor of 0.67 set in advance. 146 terms remain disputed. Across the frequency range, disagreement is not confined to rare terms: stratifying by term frequency leaves $\alpha = 0.059$ on the most common terms and $\alpha = 0.037$ on the rarest, so the disagreement is about the ladder itself rather than about thin evidence.

An α near zero can mean genuine disagreement or a skewed-marginal artefact, so we report the diagnostics that separate the two. Raw agreement is 0.530 and Gwet's AC_1 is 0.082 [19]. Raw agreement far above α with AC_1 also low is the signature of real disagreement on a hard boundary, not a paradox of the coefficient.

For that case the manual specifies a tie-break: where the evidence is inconclusive, defer to the archive's own facet depth. Deferral would ground the ladder in archive practice rather than in the annotator, and it is almost never reached, because two protocols answer the manual's questions with complementary tests that cannot return inconclusive. Only the lexical protocol defers at all, on 46.0% of terms, and the three are jointly inconclusive on 0.0%. The specification appears to settle hard cases by appeal to the archive, and in practice each implementation settles them itself.

Granularity carries the same instability. Changing it changes which cells the model is asked to rank, so we compare triage lists across the three settings after projecting them onto a common referent, without which the comparison is vacuous, since coarse and fine label spaces are disjoint by construction. Rank-biased overlap [23] between granularities is 0.537 for coarse against mid, 0.317 for mid against fine, and 0.251 for coarse against fine. All three fall below the threshold of 0.60 we set in advance.

TABLE I

A lossless star expansion is the stronger ranker at matched parameter count. Primary cell: mid granularity, narrator-disjoint split, mixed negative sampling, rank map R-A; means over 10 seeds at matched parameter budget.

model	parameters	T1 MAP	T1 nDCG@10	T2 AUC
M0 text-only MLP	130,401	0.009	0.000	0.552
M1 dense retrieval	0	0.069	0.093	0.725
M2 untyped star	130,909	0.065	0.061	0.785
M3 typed star	132,219	0.139	0.335	0.816
M4 AllSet	131,041	N/A†	N/A†	0.837
M4 ED-HNN	131,041	N/A†	N/A†	0.853
M4 Hypergraph-MLP	131,041	0.009	0.000	0.549
M5 CCNN-style (ours)	128,521	0.098	0.066	0.546

† AllSet and ED-HNN pass messages only between narrators and events and never update moment representations, which are what T1 is scored over. Their T1 values are the frozen-encoder floor, identical across both variants and all seeds, so we report them as undefined rather than as a measurement of either operator.

C. Operator advantage in the primary cell

Table I gives the primary cell. In that primary cell the typed star expansion is the strongest model on the ranking task (T1 MAP 0.139), the full complex network reaches 0.098, and a dense retrieval baseline with no learned parameters reaches 0.069. Two of the hypergraph baselines cannot be scored on T1 at all: AllSet and ED-HNN pass messages only between narrators and events and never update the moment representations that T1 is scored over, so their entries are the frozen-encoder floor rather than a measurement of either operator, and we report them as undefined. On the link task the hypergraph diffusion operator [5] is strongest (AUC 0.853), and the complex sits behind every message-passing baseline, on the input-deprived configuration whose link AUC Section VI-C raises to 0.704.

That comparison gives the star inputs the complex never receives, so the registered verdict stands: the typed star as specified beats the complex at matched parameter count, and P3 is settled against the extra structure. One run supplies all four cells of the operator-by-features factorial at 5 seeds. Carrying the spectral block the star reaches 0.147 against the complex's 0.101; stripped to identical inputs it reaches 0.065 against 0.097. On identical inputs the encodings are worth 0.147 minus 0.065 to the star and almost nothing to the complex, so the sign of the operator comparison is set by the inputs rather than by the operator. Their marginal intervals overlap ([0.049, 0.081] against [0.074, 0.122]), but the paired difference does not: resampling the 56 test narrators puts that 0.033 at [0.023, 0.042], and the encoded cells reverse at [-0.053, -0.039]. What the matched arm settles is where the advantage comes from: it grows to 0.153 once narrators straddle the split, so most of it is the exposure this paper is about. Encodings are not wasted on the complex, since its link AUC rises from 0.550 to 0.704; it cannot route them to the rank the retrieval task scores. The gap is 1.4-fold in MAP and 5.1-fold in nDCG@10, so the complex places relevant passages somewhere in the list but rarely near the top, the region an archivist actually reads.

In Table II, the first ablation removes the moment rank and

TABLE II
REMOVING THE MOMENT RANK COLLAPSES RANKING AND IMPROVES LINKING, SO THE TWO TASKS DO NOT WANT THE SAME STRUCTURE. ABLATIONS OF THE COMPLEX NETWORK, PRIMARY CELL.

ablation	T1 MAP	T2 AUC
share weights across rank pairs	0.093	0.557
shuffle the rank assignment	0.063	0.544
collapse ranks 2 and 3	0.036	0.558
remove down-messages	0.052	0.544
remove the moment rank	0.008	0.840
remove within-rank exchange	0.076	0.543

raises link AUC from 0.546 to 0.840. Ranking for that ablation is not interpretable: T1 is scored over moment representations the ablation removes from the ladder, so its collapse is guaranteed by the metric.

Link results stand on their own, and the complex's is not an optimisation failure: with dropout off it drives training loss to 0.000 on 200 held-in incidences while its validation AUC never exceeds 0.545 at any of 600 epochs, and its best anywhere on the tuning grid is 0.568. The ceiling is representational. Nor does it move under the selection rule or with tuning: reselecting by validation MAP widens the narrator-disjoint gap from -0.050 to -0.061 over 5 seeds, and an equal-budget grid of 8 configurations per model over 3 seeds leaves -0.050, with the complex's best T1 MAP, 0.103, below the star's worst, 0.133. The star's winning configuration on that grid is the registered default, lr = 0.005 and dropout 0.1, so no tuning advantage was withheld from the baseline the argument runs against. All three architectures that never update moment representations, this ablation and the two rank-agnostic baselines, sit at the top of the link range, while the complex that keeps the moment rank sits near chance. The moment rank is harmless in itself, since the star expansions also carry it and score well above the complex. What the ablation isolates is narrower: inside this complex, the intermediate rank is where the rank-aware operator loses the link task.

Two further checks bound the point estimates. Both start from the unit of dependence: narrators, whose design effect is 139.0 (ICC = 0.169 over 470 narrators), so intervals computed as though the 383,680 incidences were independent would be narrower than honest ones by roughly 12. On the link task the complex reaches a narrator-clustered per-incidence accuracy of 0.517 (95% CI [0.499, 0.536]), an interval containing chance. Narrator counts differ by statistic, each naming a population: 470 in the archive, 447 with a per-narrator gap, 440 probeable, 384 holding rank-2 incidences, 188 bearing test queries.

Our second check is sharper. Predictions enter it averaged over the 10 seeds within each held-out incidence before fitting, so seeds do not enter as independent observations; the model then regresses correctness on architecture, log event size and their interaction, with crossed random intercepts for narrator and event. Fitted over 75 events and the 445 narrators appearing in test-set pairs, most of them through sampled negatives rather than through a held-out incidence, it estimates the interaction at -0.085 (95% CI [-0.104, -0.065], $p = 2.5 \times 10^{-17}$). Event count here is smaller than the 141 rank-2 cells in the complex because the model is fitted on held-out test incidences. Fitted that way, the interval excludes zero, and the sign is the one the representation should have ruled out: the complex falls further behind the typed star as events grow, precisely where the extra ranks were supposed to earn their place.

This coefficient is fitted on the primary cell alone; the release reports the pooled estimate, -0.085 over 75 events, and why mixing three granularities with the random split measures nothing the paper claims.

Under the test fixed in advance, almost stochastic order [24] Bonferroni-corrected across 110 model pairs and computed with deep-significance [25], the typed star stochastically dominates the complex on T1 ($\varepsilon_{\min}$ = 0.00), while the reverse comparison returns $\varepsilon_{\min}$ = 1.00, the maximum possible violation. The correction runs over all 110 ordered pairs rather than the single pair P3 names, which leaves it conservative here.

One defence of a rank-structured model remains: it should pay off precisely where structure is richest, on events many narrators describe. Panel (b) of Figure 2 tests that directly and finds the opposite: the complex leads only on the two smallest bins, where the intervals are widest, and falls progressively further behind as events grow, the same pattern the interaction estimate reports.

VII. SPLIT DEPENDENCE OF THE MODEL ORDERING

Panel (c) of Figure 2 was not planned in advance. Under exactly one of the three splits, the pre-registered narrator-disjoint one, the typed star leads. Under a random split the ordering inverts and the complex leads by a wide margin (0.388 against 0.106). Under an event-disjoint split it leads again. What those splits share is the partition rather than the architecture: only the narrator-disjoint split keeps a narrator's segments on one side, so the other two reward a model for recognising a narrator it was trained on.

The quantity carrying the claim is the difference between those two gaps. Seeds measure optimisation noise, whereas the sampling unit the claim is about is the query, and queries cluster by narrator, so we resample 188 narrators with replacement over 10,000 replicates and average over seeds within each. Across those replicates the complex's random-split gain exceeds the star's by 0.295 MAP, with a 95% interval of [0.258, 0.331]. That estimate resamples query-bearing narrators and averages inside each replicate, so it does not equal the difference of the cell means above. Nor does the reversal depend on the selection rule: reselecting every checkpoint by validation MAP leaves it at 0.236 against 0.239 over 5 seeds.

Little else matters. Table III varies every design axis the field argues about, granularity, negative-sampling regime and rank map, and the ordering survives 9 of 11 conditions. One axis reverses that ordering, and it is usually a default rather than a decision.

The random split is a common default, and on this archive the two splits support opposite conclusions. A model's score under a random split minus its score under an entity-disjoint one is its *entity exposure gap*, which panel (c) of Figure 2 reports per model.

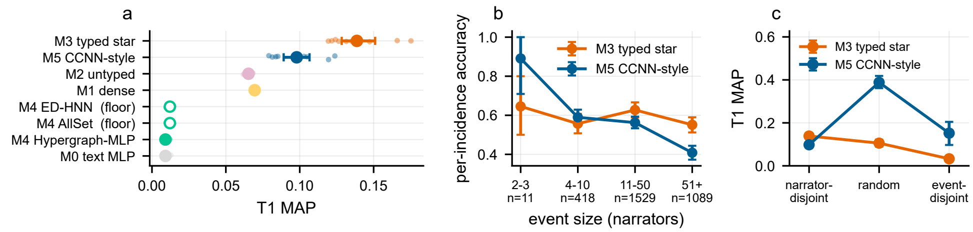

Fig. 2. The typed star leads at matched budget, its margin widens as events grow, and the evaluation split decides the ordering. (a) T1 MAP in the primary cell with all ten seeds shown individually, because means hide bimodality. Hollow markers labelled *floor* are AllSet and ED-HNN, which never update rank-1 cells, so their T1 values are the frozen-encoder floor. (b) Per-incidence accuracy stratified by event size with narrator-clustered 95% intervals, where n counts held-out incidences. (c) The same two models under all three splits with everything else held at the primary cell; both splits that let a narrator appear on both sides of the partition favour the complex. Intervals in (a) and (c) are bootstrap 95% CIs over ten seeds, which measure optimisation noise; the narrator-clustered interval the split claim rests on is in the text.

TABLE III

THE SYSTEM-LEVEL ORDERING SURVIVES EVERY DESIGN CHOICE EXCEPT THE SPLIT. EACH ROW VARIES ONE AXIS WITH THE REST HELD AT THE PRIMARY CELL, AND REPORTS HOW OFTEN THE TYPED STAR LEADS THE COMPLEX ON T1 MAP. EVERY CELL USES THE STAR AS REGISTERED; THE INPUT-MATCHED ORDERING IS IN SECTION VI-C.

design axis varied	levels	typed star leads
granularity	coarse, mid, fine	3 / 3
negative sampling	MNS, UNS, SNS, CNS, hard	5 / 5
rank map	R-A, R-B, R-C	3 / 3
split	narrator-disjoint, random, event-disjoint	1 / 3

Per-axis counts share the primary cell, so they do not sum: over the 11 distinct cells the typed star leads in 9.

VIII. MECHANISM: NARRATOR IDENTITY IN THE FEATURES

An explanation is owed, and two experiments run after the split result, labelled exploratory, supply one more concrete than idiolect. Feature construction is identical for every model: a rank-0 narrator is the mean of that narrator's segment embeddings, and a rank-k cell is the mean of its constituent rank-0 features. Features are built once from the whole archive and split afterwards, so a narrator's vector aggregates every segment that narrator contributed, and the narrator-disjoint split closes both routes at once, the repeated vector and the cross-partition text. Those routes exist because the obstruction of Proposition 1 reaches the features, and that consequence does not depend on this archive.

Proposition 3 (Lifting carries the obstruction into the features): Let rank-k features be lifted from rank 0 by a permutation-invariant multiset reduction φ, so that a cell's input is $\varphi(\{x_s : s \in \mathrm{supp}(c)\})$. Cells sharing a support share an input, and the distinct rank-1 inputs number at most the distinct supports. Where a cell has the single owner $\pi(c)$, its input is a function of $\pi(c)$ alone, and separating two such cells requires message passing through a cell of rank at least two.

So 13,315 moments admit at most 455 distinct rank-1 inputs, a 29.3-fold collapse (Figure 1b). For the mean, $\varphi(\{x\}) = x$, so a moment's input vector *is* its narrator's vector; a learnable φ changes the value but not the dependence on $\pi(c)$ alone. We verified the equality: for every narrator holding more than one such moment, the largest elementwise difference between two of them is exactly zero.

The split decides whether that degeneracy is visible. Of 384 narrators, 249 straddle the random split (64.8%) and 235 the event-disjoint one (61.2%), since holding out events does not hold out the narrators who witnessed them. Both place a narrator's vector on both sides of the partition. A narrator-disjoint split places none. Both rates are predictable before any model exists: the random split partitions incidences, so assigning a narrator's k of them independently across folds of proportion p_f predicts $1 - \sum_f p_f^k$, or 64.1% against 64.8% observed, with the 68 single-cell narrators contributing zero. Exposure is not an artefact of one lifting: across 9 liftings the singleton fraction ranges 0.140 to 0.654 while predicted exposure stays 0.599 to 0.735, because it follows from how many cells an entity owns rather than from how many it owns alone. Those liftings are this archive's rank maps and granularities, so the sweep bounds the choice rather than the collection.

That first experiment is easy to over-read, so its control comes first. A linear probe recovers narrator identity from the *input* rank-1 features at 1.000 over 440 narrators and 12,846 moments, against a majority-class floor of 0.011, a tautology since the classes are identical vectors. That probe is a multinomial logistic regression over standardised rank-1 vectors, fitted on a stratified 70/30 split of segments held fixed across seeds and restricted to narrators with at least four segments; it partitions segments rather than narrators, so every class appears on both sides. What the trained models show is how much inherited identity survives message passing, and the answer is most of it: 0.902 for the complex against 0.862 for the typed star under the narrator-disjoint split, 0.950 against 0.920 under the random one. Complex scores are higher in both and the seed spreads overlap, so neither architecture discards narrator identity.

A second experiment intervenes on the mechanism: we shuffled narrator labels across segments, preserving segments per narrator, and re-ran both splits. Anonymisation changes which passages count as positives, so an anonymised score is not comparable to a real one; what is comparable is the *gap between splits*, because both are transformed alike. With real narrators the complex gains 0.290 MAP moving from the narrator-disjoint split to the random one. With identity destroyed that difference is -0.006, a reduction of 0.296, so its whole advantage disappears. Its advantage gone, the star's gap falls too, from -0.033 to -0.165, by less than half as far and from a value that was never positive: the shuffle costs the complex its advantage and costs the star a gap it did not have.

Shuffling also separates the two routes: it preserves how many segments each label holds, so cross-partition aggregation survives while identity correspondence does not. That reading carries a boundary, since a shuffled aggregate has less to leak, so computing features split-first is the direct test.

Each narrator's vector is rebuilt from that narrator's training-side segments, a segment counting as training-side when none of that narrator's incidences through it are held out, with a fallback to a narrator's own segments that is not a leak because a new interview arrives at deployment with its transcript. Under the random split the construction strips a mean 13.8% of the segments contributed by each of 266 affected narrators.

The reversal survives it, which is what we predicted in writing beforehand. Moving from the narrator-disjoint split to the random one, the complex gains 0.268 MAP under global features and 0.306 under split-first ones, and the input-matched star gains 0.165 and 0.163. Both arms come from one re-run of the primary cell, so they are comparable to each other rather than to the grid values above. Removing every token of test-side text leaves the advantage where it was, so cross-partition aggregation is not the carrier and what the random split rewards is repeated narrator identity.

The spectral encodings suppress exposure in both architectures, which is the second lever on the same mechanism. Across the factorial, the star's exposure gap falls from 0.177 on plain inputs to -0.031 with the encodings, and the complex's from 0.297 to -0.025. On matched inputs both leak, the complex by roughly 0.297 against 0.177, so the leak is differential rather than present in one architecture and absent in the other. Features that carry structure rather than identity reduce what a random split can reward, whichever operator reads them.

The random split does not merely correlate with the complex's advantage: removing what it leaks removes the advantage. Because a moment is represented by its narrator, a random split places *the same input vector* on both sides, and the operator that propagates further through the narrator layer is the one the default split rewards.

A third experiment bounds that account. If the leak were carried by the moment vector alone, replacing it should close the gap, so we gave every rank-1 cell its own passage embedding instead of its narrator's mean, holding the complex, the splits, the seeds and the budget fixed. Two moments by one narrator are then no longer identical at the input, and the gap still did not close. Against that experiment's own baseline it moved from 0.315 to 0.385 for the complex, slightly wider rather than narrower, and the typed star stayed where it was (-0.024 against -0.018).

We report that as a refuted prediction, because it bounds the claim. Only the rank-1 features changed; the rank-0 narrator layer and every message path through it stayed in place, and that is what anonymisation destroys, which is why splitting on the narrator fixes the leak and editing features does not.

A dose-response prediction is also on the record: the complex's per-narrator random-split advantage should rise with that narrator's segment count, against a flat control slope. Over 447 narrators the random-split slope is 0.015 (95% CI [-0.010, 0.040]) against a control slope of -0.169. That control is not flat: the complex falls further behind on prolific narrators exactly where it cannot have seen them, so the mechanism stands on the intervention rather than on the correlation.

Our dense baseline is scored on each passage's own frozen embedding while every trained model sees its narrator's mean, so 0.069 reflects that trade; the typed star against the complex, which is what the paper claims, sits on identical inputs.

IX. Four rival explanations

A negative result is worth reporting only if it survives the obvious explanations. Four were tested; the segmentation check settles at once, since its three defensible extraction settings are the granularities of §VI-B renamed, agreeing no better than 0.251.

a) The ordering holds on a cleaned grid.: The event layer is under-merged (§IV), so perhaps the complex is being asked to model damage. Rebuilding the archive with the 21 contested terms removed leaves 4,538 segments and 127 rank-2 cells, against the 141 the primary complex carries. All models rise on the reduced grid because the candidate pool shrinks. On that grid the star–complex ordering is unchanged, 0.127 against 0.091, so the complex is not being asked to model damage. A parameter-free baseline rises furthest, to 0.132, and overtakes both, which is a pool-size effect and a reminder of how little of this ranking the learned structure contributes.

b) Curated cells survive their own error rates.: We propagated the measured merge and split error rates through the complex over a 25-point grid per calibration, twenty draws each. Calibrated to the archive's internal disagreement, the triage list retains RBO_{50} = 0.978 against the unperturbed ranking, clear of the registered threshold of 0.70, and that is the figure governing the triage we report, since the complex analysed here is built from the curated vocabulary. Substituting the automatic extractor's own error rate drops retention to 0.324, below the threshold, whose committed consequence is that an automatically built triage list is a screening step needing human verification. Instability here is inherited from the extractor, not from the curated cells.

c) Counting interviewers erases the phenomenon.: Counting interviewer turns as attestations would inflate every cell: the singleton fraction falls from 0.206 over narrators to 0.000 once interviewers count, because an interviewer is present in every segment, so every event acquires a second voice and nothing is under-attested. Both triage lists still agree at 0.882, which is why the decision looks harmless until the quantity that matters is inspected.

X. Utility of the representation

The application claim is archival triage: surfacing events that rest on too few voices. The quantity ranked is *archive-conditioned* attestation multiplicity, how many narrators *in this*

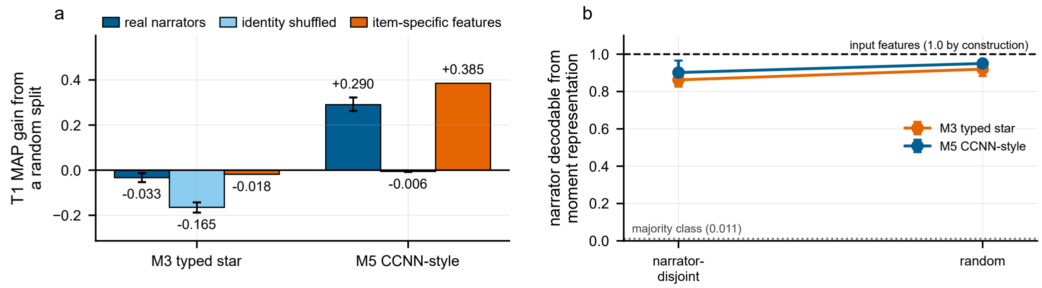

Fig. 3. Destroying narrator identity removes the complex's random-split advantage, and editing the moment features does not. (a) Gain in T1 MAP from switching to a random split, with 95% intervals over seeds. The item-specific arm was run separately and is measured against its own baseline, so its bar carries the same quantity but not the same reference as the other two. Shuffling narrator labels takes the complex's gain to zero; replacing each moment's vector with its own passage embedding leaves the gain intact, which locates the leak in the narrator layer rather than in the moment vector. (b) A linear probe recovering narrator identity from moment representations, against a majority-class floor of 0.011 over 440 narrators. The dashed line is the same probe on the input features, 1.000 by construction. The axis runs the full range.

collection describe an event, so an event thinly attested here may be abundantly documented elsewhere, and it presupposes curation, since rank-2 cells come from the archive's controlled vocabulary. Two questions decide whether the machinery earns its place.

a) A minority of triage items are unstable under resampling.: Of the top 50 triage items, 8 are unstable under the conditions tested. The list is a screening device, and each item ships with the probability that it is a singleton only because of extraction error.

b) Mention frequency sets the bar the structure has to clear.: Archive-conditioned attestation multiplicity correlates with bare mention frequency at $\rho = 0.879$, the two triage lists agree at $\mathrm{RBO}_{50} = 0.444$, and they share a Jaccard overlap of 0.500 in their top ten. Counting mentions recovers most of what the ladder, the encodings and the complex were built to recover, so the structure-derived ranking has to justify itself where the two disagree, and that region is the minority of the list.

XI. Transfer to a held-out collection

What transfers is a conclusion rather than a model: every threshold, rank map and hyperparameter was applied as fixed on the primary archive, with no tuning, to held-out collections of 55 interviews and 874 segments. Rank and cardinality: $\rho = 0.457$ over 32 rank-2 cells. Rank agreement: $\alpha = 0.062$. Models: typed star 0.356, complex 0.347. Those levels are not comparable with the primary archive's, since this collection carries 32 rank-2 cells against 141 at home and has no transcripts, so we leave the model comparison unread there. Our pre-registered test is narrower: whether the interaction estimate keeps its sign and magnitude. It does not, so the interaction finding is archive-specific, as we committed in advance.

XII. Release and instruments

A registered topology analysis is absent, and the reason generalises: its permutation null tests nothing, since shuffling which rank-2 cell holds which narrator set is a relabelling and homology ignores labels, so $p = 1.000$ was forced by construction. Event-size and connectivity moments predict β_1 at $R^2 = 0.998$, above the registered 0.9, so the verdict is no.

Every estimator was asserted against a known answer before being pointed at the archive, which caught eight defects that would otherwise have been reported as findings, among them a link task whose message passing was handed the edges it had to predict. Every reference was retrieved programmatically and accepted on a title match above 0.92 with the first author confirmed. Our release, *CAIRN* (https://github.com/AlexDongzeyu/CARIN), is stand-off because the archive's material is not ours to redistribute, and carries the cell definitions, incidence matrices at all three granularities, the splits, the annotation manual, a licence audit, the defect log, the substitution record and a fetch script. A fresh interpreter recomputes the headline corpus statistics from those files alone, which is the reproduction the release supports; independent human replication is the one substitution still outstanding.

XIII. Limitations

Six substitutions are on the record, each bounding the claim in a stated direction. The agreement study measures whether the manual determines the assignment rather than whether people agree, and the transfer collection refutes a strong effect without confirming a weak one. Four others are registered: no expert panel, no external coreference corpus, no second archive, and a self-sufficiency check for independent reproduction. No contemporary generalised-CCNN baseline was run, so the operator comparison is with the star expansion and the hypergraph family rather than with that line. Our corpus is one archive in one language, and the ladder is its cataloguers' choice, so a different descriptive tradition could behave differently. What survives those caveats is the structure of the argument: the three preconditions are testable anywhere, and the lifting creates the same exposure wherever cells share recurring owners, though the magnitude and the reversal are findings on this archive.

XIV. Conclusion

The ownership obstruction is what this archive turned up. Where items are singly owned, the cells above the owners are not subsets a combinatorial complex can hold apart, and here 13,315 moments collapse onto 455 supports, so a lifted rank-1 feature is a function of who owns it. Rank carries information beyond cell cardinality and the star expansion is lossless, so the ladder is real; the specification does not yet supply a reproducible assignment, and the complex has still to earn its extra structure. What the star wins with, though, is its inputs: on matched features the ordering reverses, so the registered verdict is about the system rather than the operator.

Evaluation decides whether the collapse is visible. A random split lifts the complex 0.295 MAP more than it lifts the star (95% CI [0.258, 0.331]), and that difference survives rebuilding every feature split-first, so the carrier is repeated narrator identity. Shuffling that identity removes the advantage, and giving each moment its own passage text does not. Beyond this archive, the obstruction reaches any collection whose items belong to recurring entities, so the defence is to split on the entity, and to check what a rank-1 cell contains before concluding that rank helped.

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